

Introduction to LaTeX

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What is Latex?

A typesetting system used to produce professional looking documents[1]

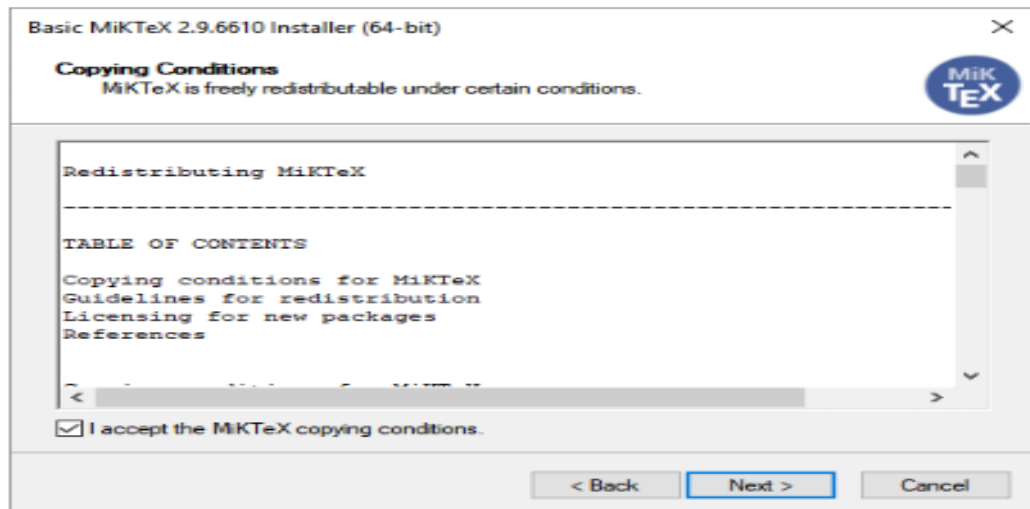
Why LaTeX?

- It is good at dealing with mathematically oriented manuscripts.
- Can be used to produce a variety of document types, including presentations, articles and reports.
- It is good at management of a bibliography.

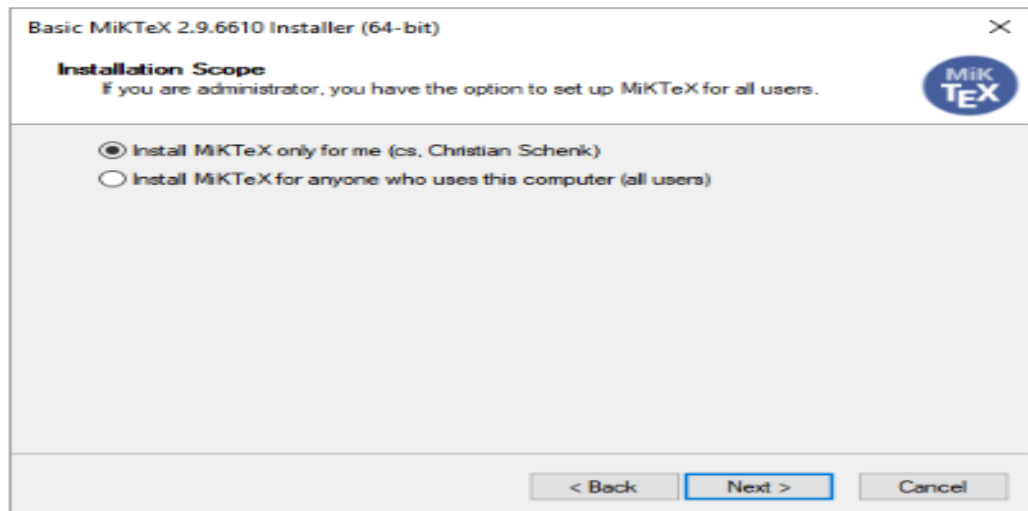
Installation

- MiKTeX download: <https://miktex.org/download>
- TeXworks download: <https://softtradar.com/texworks/>

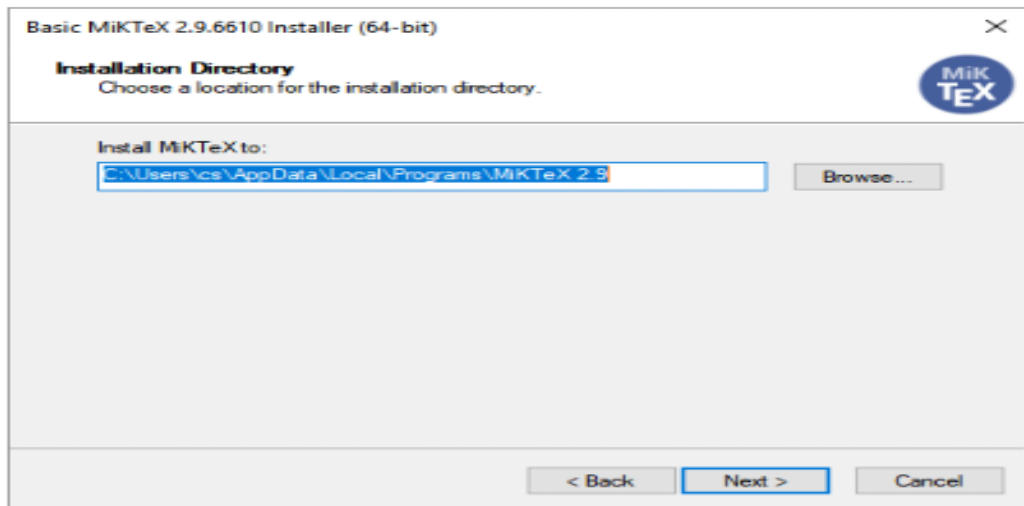
MiKTeX installation step 1



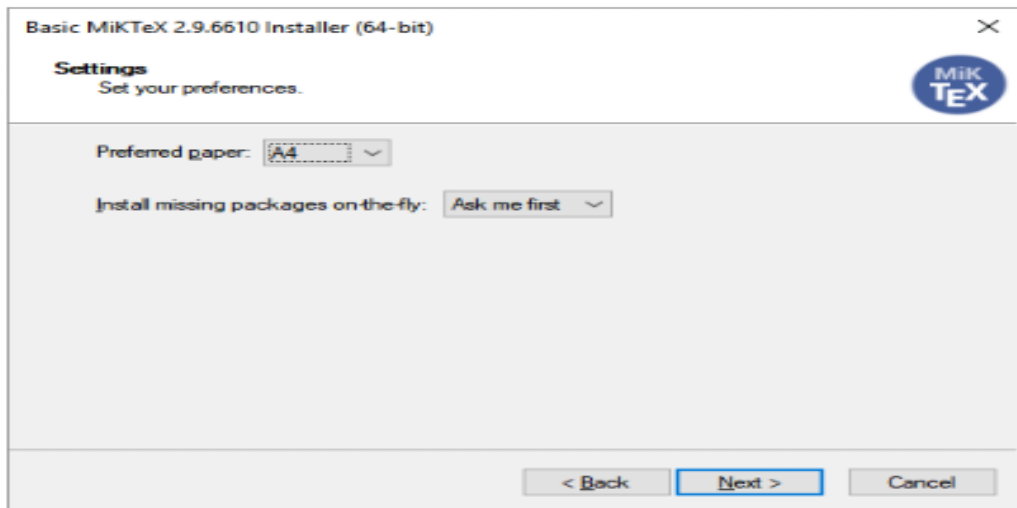
MiKTeX installation step 2



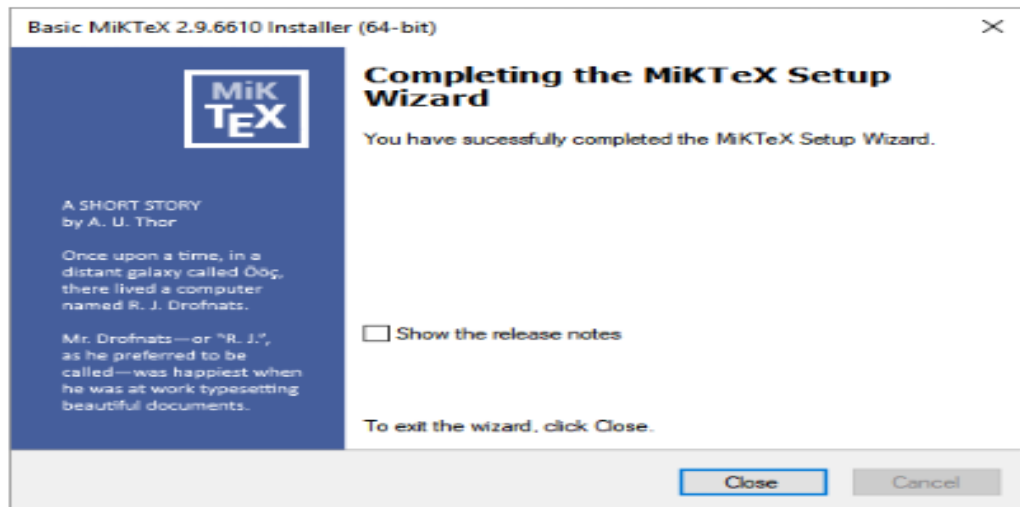
MiKTeX installation step 3



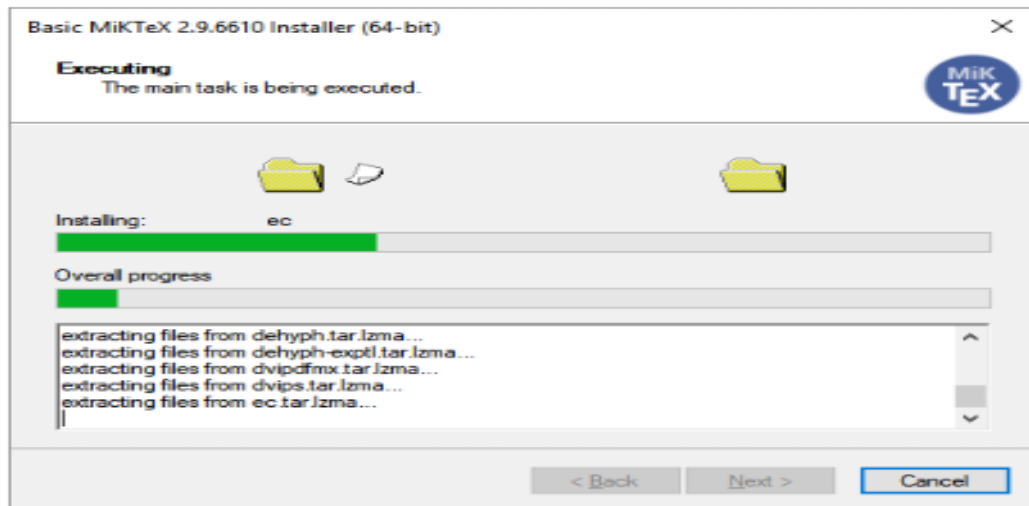
MiKTeX installation step 4



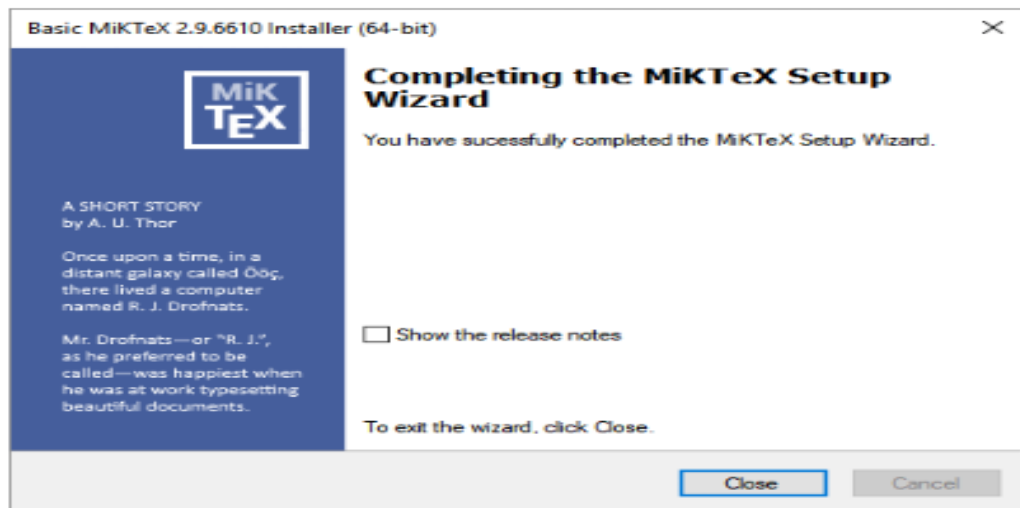
MiKTeX installation step 7



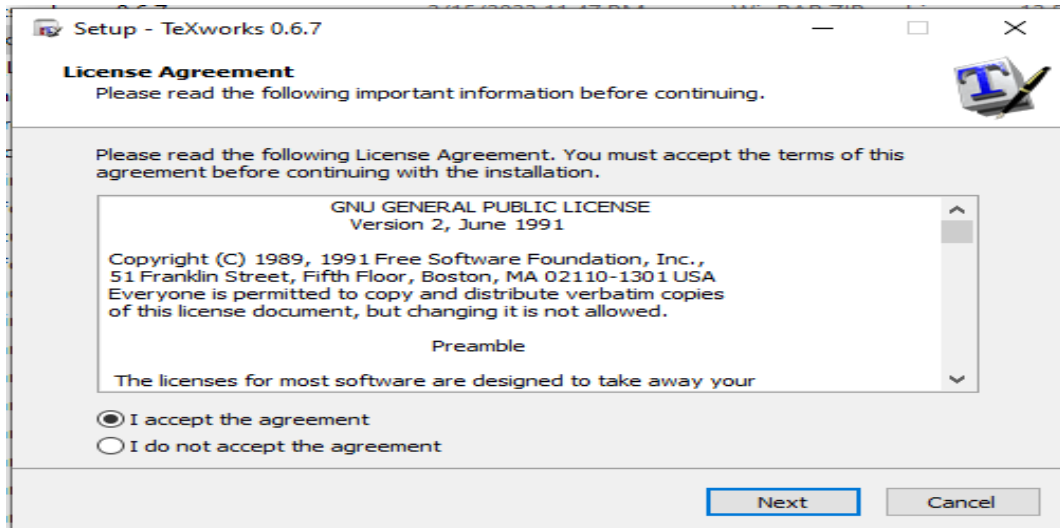
MiKTeX installation step 6



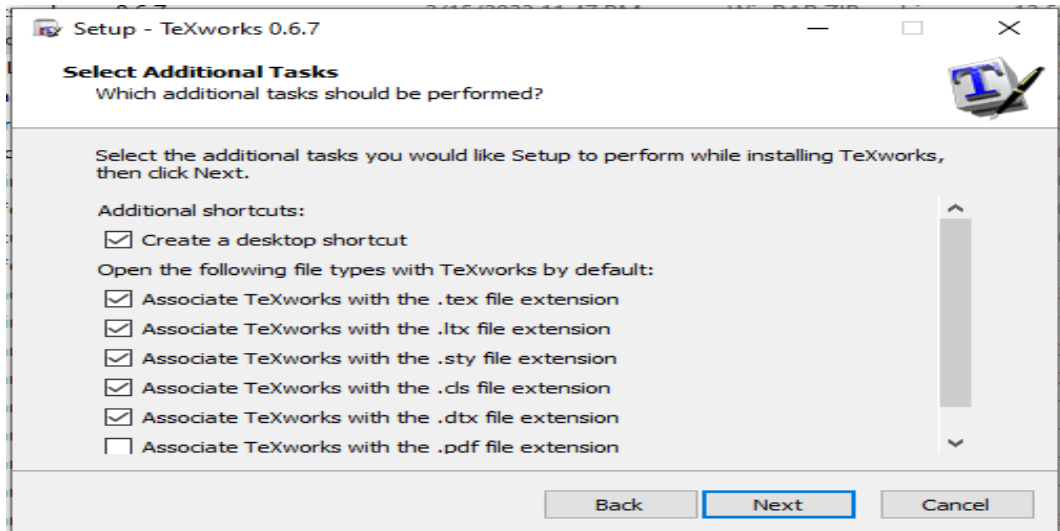
MiKTeX installation step 7



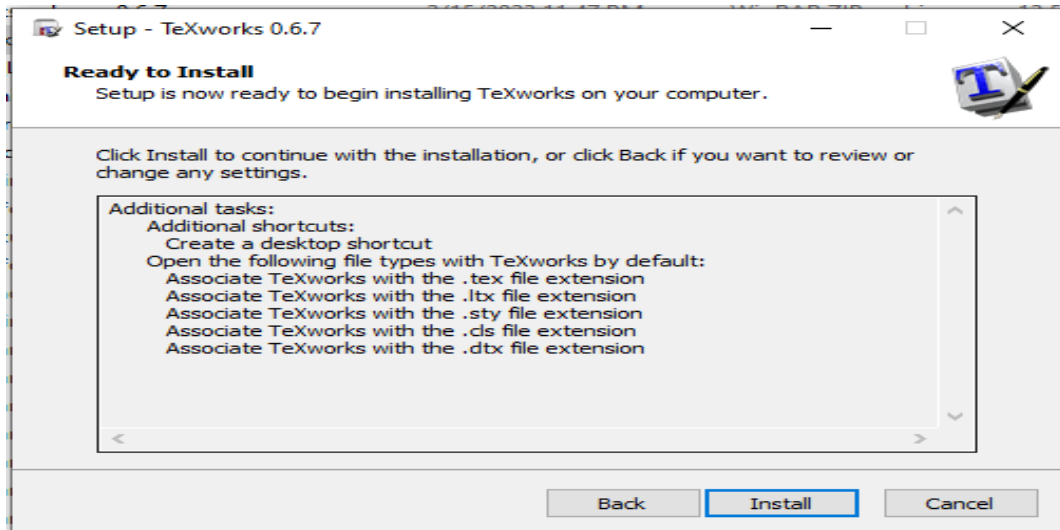
TeXworks installation step 1



TeXworks installation step 2



TeXworks installation step 3



Structure

```
\documentclass{...}
```

```
\usepackage {...}  
\begin{document}
```

```
.....
```

```
\end{document}
```

Structure cont'd

The area between `\documentclass{...}` and `\begin{document}` is called the preamble. It normally contains commands that affect the entire document. After the preamble, the text of your document is enclosed between two commands which identify the beginning and end of the actual document.

documentclass

`\documentclass[options]{class}`

- LaTeX needs to know the type of document the author wants to create.
- class specifies the type of document to be created.
- The options parameter customizes the behaviour of the document class

Main LaTeX document classes and their use

latex document classes	Usage
article	To produce articles for scientific journals, short reports, program documentation, and, in general, any kind of short document that doesn't need a complex division into chapters and parts.
report	A kind of long article that can contain several chapters. It is usually used for small books and thesis.
book	Used to typeset real books, includes all the kinds of typesetting standards that one may expect from a printed book: chapters and parts, two kinds of pages, a front matter, etc
slides	It was used to produce slides, although the external package beamer  is much more powerful for this purpose.
letter	For writing letters.

Example

- `\documentclass[12pt, twocolumn]{article}`
- `\documentclass[12pt,oneside]{report}`
- `\documentclass[12pt]{book}`
- `\documentclass{beamer}`

Options

- Font size (10pt, 11pt, 12pt)
- Paper size and format (a4paper, letterpaper, etc.)
- Draft mode (draft)
- Multiple columns (onecolumn, twocolumn)
- Formula-specific options (fleqn and leqno)
- Landscape print mode (landscape)
- Single- and double-sided documents (oneside, twoside)
- Titlepage behavior (notitlepage, titlepage)
- Chapter opening page (openright, openany)

Packages

A comprehensive list of packages and documentation for them can be found on : <https://ctan.org/>

- ① `\usepackage{xcolor}` : Allow colors to be defined
- ② `\usepackage{graphicx}` : Essential for inserting figures
- ③ `\usepackage{hyperref}` : Makes citations “clickable” in the .pdf document

Title Page

```
\title{Introduction to LaTeX }\\
author{David Hagumuwumva\\
Faculty of IT,\\
Adventist University of Central Africa\\
\texttt{haguma@auca.edu}\\
}\\
\date{\today}\\
\maketitle
```

Abstract

```
\documentclass{article}  
\begin{document}  
\begin{abstract}  
Your abstract.....  
\end{abstract}  
\end{document}
```

Section

```
\documentclass{article}
\begin{document}
\section{Introduction}
Some text here...
\section{Covid-19 trend}
Type here .....
\subsection{Global death rate}
\subsubsection{Death rate in Rwanda}
\subsection{New cases}
\end{document}
```


TOC, LOF, LOT

Once you have your sections set up correctly you can then include a table of contents in your work. To do this simply insert the following command where you would like your contents to appear in the document.

```
\tableofcontents
```

You are also able to include a list of figures or tables that you have specified in the document in a similar way using the following commands.

```
\listoffigures
```

```
\listoftables
```

List

Bulleted List

```
\begin{itemize}
\item Item 1
\item Item 2
\end{itemize}
```

Numbered List

```
\begin{enumerate}
\item Item 1
\item Item 2
\end{enumerate}
```

Description List

```
\begin{description}
\item[First] Item 1
\item[Second] Item 2
\item[Second] Item 3
\end{description}
```

Nested Lists

```
\begin{enumerate}
\item Item 1
\begin{enumerate}
\item Nested item 1
\item Nested item 2
\end{enumerate}
\item Item 2
\end{enumerate}
```

Labels and Referencing

```
\section{Introduction}
```

```
\label{sec: intro}
```

In this section we

```
\section{Referencing}
```

Now we can refer to section `\ref{sec:into}`.

Table

```
\begin{tabular}[pos]{table spec}
```

```
... table here ...
```

```
\end{tabular}
```

More details on table:

<https://www.overleaf.com/learn/latex/Tables>

Math

Inline and displayed formula are two basic types of formulas that can be used in LaTeX. Inline are defined and displayed in the text you type while displayed are separate from the main text.

`\usepackageamsmath`

inline example: $a + b = c$

Displayed example:
$$a + b = c$$

For more details visit:

<https://en.wikibooks.org/wiki/LaTeX/Mathematics>

Mathematical expressions

Please visit: <https://en.wikibooks.org/wiki/LaTeX/Mathematics>
and
https://en.wikibooks.org/wiki/LaTeX/Advanced_Mathematics
for mathematical expressions

Graphics

```
\usepackage{graphicx}  
\includegraphics{image.png}  
\includegraphics[scale=0.7]{image.png}  
\includegraphics[width=2.7cm]{image.png}
```

For more details visit:

https://en.wikibooks.org/wiki/LaTeX/Importing_Graphics

Bibliography

There are two main ways to include a bibliography in your work. You can store them in an external BibTeX file and then link to them via a command to your current document. This way you can create a file of your references that you can simply link to them when you need. However for single documents it is much simpler to embed the references in the file itself.[?]

Bibliography

```
\begin{thebibliography}{20}  
\bibitem{Ma} Martin D. Crossley, \emph{Essential Topology},  
Springer-Verlag London Limited 2007.  
\bibitem{Co} Colin Adams and Robert Franzosa, \emph{Introduction  
to Topology: Pure and Applied}, Pearson Education, Inc, 2007.  
\end{thebibliography}
```

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Generate the first project

Note: The majority of future project reports and thesis should be printed in LaTeX.

Bibliography



Tobias Oetiker, Hubert Partl, Irene Hyna, and Elisabeth Schlegl.
The not so short introduction to latex2 ϵ .
Bulgarian, Chinese, Czech, Dutch, Estonian, 1995.